NOVA IMS
INFORMATION MANAGEMENT SCHOOL

“What makes a music artist successful?”

Using data to understand what makes artists popular on major music streaming platforms

Programming for Data Science 2026/2027
Group Project



1. Project Context

Music streaming platforms generate large-scale data about listening behavior, popularity trends, and artist impact. Your task is to transform a raw “All-time most streamed artists” dataset into an analysis and/or tool that helps someone answer practical questions such as:

- Which artists dominate streaming?
- Where are they from?
- Are there identifiable patterns by genre?
- Can we build an exploration or recommendation-style insight based on the dataset?

You will work through a full Data Science workflow: loading and cleaning the dataset, exploring it, visualizing it (with static and interactive visualizations), and tackling advanced tasks (e.g., web scraping for enrichment or geo/image features usable from external sources).

2. Project Objectives

By the end of the project, you will be able to:

- Develop a comprehensive understanding of the steps required to start an exploratory Data Science project from scratch.
- Demonstrate proficiency in importing, cleaning, manipulating, and wrangling datasets using Python.
- Apply exploratory data analysis (EDA), web scraping, and visualization techniques.
- Produce interpretable insights and connect results back to the original context.
- Handle multiple data types and prepare results for downstream analysis.
- Apply good programming practices: readable code, modular organization, efficiency, and reproducibility.

3. The Dataset

This dataset breaks down all-time streaming metrics for the world's most popular musical artists on a streaming platform. As of 2026, this data contains information on the reach of artists across various genres, countries, and types (Solo vs. Group).

The data highlights the nuance of streaming success, not just total volume, but the split between "Lead Streams" (where the artist is the main credit) and "Feature Streams" (where they appear as a guest), providing a clearer picture of an artist's market versus their collaborative reach.

The dataset includes:

Variable	Description
Artist ID	Unique identifier for each artist
Artist Name	Name of the artist
Sex	The Sex of the Artist
Country of Origin	Country associated with the artist
Primary Language	Main language associated with the artist
Primary Genre	Main musical genre associated with the artist
Artist Type	Whether the artist is classified as a solo artist or group
Debut Year	Year in which the artist debuted
Update Date	Date when records were updated
Year	Year when the dataset was extracted
Total Streams	Total streams, in millions
Lead Streams	Streams where the artist is the main credited artist, in millions
Feature Streams	Streams where the artist appears as a featured artist, in millions
Solo Streams	Streams associated with solo work, in millions
Collaborative Streams	Streams associated with collaborative work, in millions

Note that Total Streams can be broken down in two perspectives:

- Total Streams = Solo Streams + Collaborative Streams

- Total Streams = Lead Streams + Feature Streams

4. Project Requirements

4.1. Data Wrangling and Analysis

You must cover the core Data Wrangling and Analysis topics below by answering all the questions below.

1) Are there duplicates or missing values in the dataset? If so, how did you handle them? Justify your decisions. Are there any other data quality issues that you found? If so, what are they and how did you handle them?

2)

- **a)** In total, how many unique artists are included in the data? How many pop artists are there?
- **b)** What is the most common primary genre for artists? How many artists are primarily associated with that genre?
- **c)** What is the most common country of origin among artists? How many artists are from that country?
- **d)** What are the top 3 most common primary languages among artists? Produce a plot that allows you to answer this question.

3)

- **a)** How many days ago were the most recent and oldest updates to the data? On what day, month and year did they occur?
- **b)** Are there any artists with out-of-date stream counts, i.e. counts whose last update was before the end of 2025? If so, which artists require an update to their stream counts?

4)

- **a)** How does the length of an artist's career impact their total streams? What can you conclude from it? Provide visual representation.
- **b)** What are the top 5 artists that benefit the most from collaborations, having a larger % of their streams coming from collaborations? Support your claim with a visualization.

5) What are three additional insights you can uncover from the dataset (e.g. when comparing artists and the impact of different factors on stream counts)? When relevant, support them with visual elements.

4.2. Advanced Topic - Building Interactive Maps

The goal for this section is to create interactive maps of the world that users can use to explore stream trends globally and view relevant information, such as the language, genre, and stream counts of artists. This encompasses two main tasks:

1) **Web Scraping**
   - Starting from the Wikipedia Main Page (https://en.wikipedia.org/wiki/Main_Page) extract the birthplace or origin of artists, then extract the coordinates of those locations.
   - Locations and coordinates must be obtained only through this method
   - Ensure that artists are correctly matched to their birthplace or origin and that the coordinates are correctly matched to the corresponding locations.

Note that, on Wikipedia, it may not be possible to find a birthplace (or origin) for some artists or coordinates for some of the locations. However, exploring how to reach as many as possible within Wikipedia will be rewarded.

2) **Interactive maps**
   - Using the scraped coordinates build, at least, two maps:
     - One world map where each marker represents an artist and should be sized according to the total number of streams.
     - A density world map showcasing the global distribution of streams based on the birthplace or origin of artists.
   - For the normal world map, each artist marker should display the following information when hovered or clicked on:
     - The artist's name
     - The birthplace or origin of the artist
     - The total number of streams of the artist
     - The % of solo and collaborative streams of the artist
   - You may use any tools covered in class (e.g., plotly, geopandas, etc).
   - Below are some baseline examples. However, you are free to do more graphs or customize the design freely as long as the required information is provided.

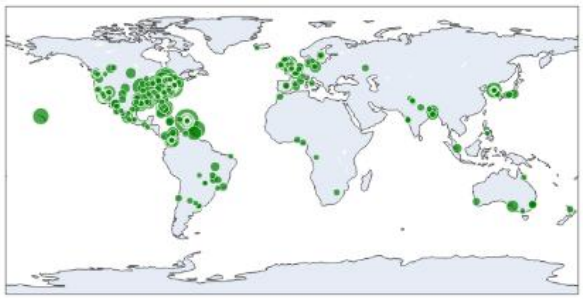


Figure 1 - Example of World Map

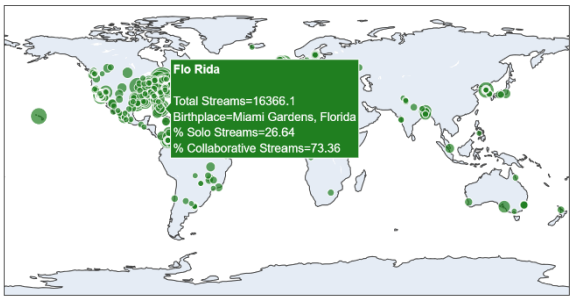


Figure 2 - Example of hover for an artist (Flo Rida)

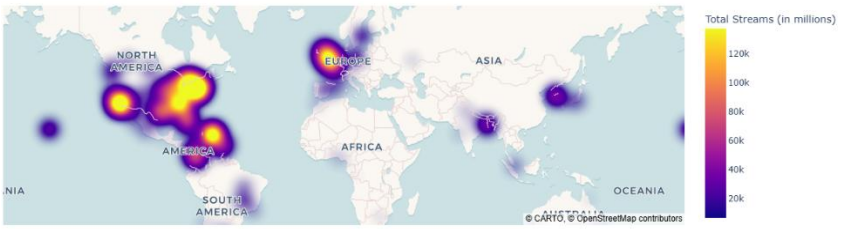


Figure 3 - Example of Density Map

4.3. Data Science in Action

In this final phase, you will bring everything together by integrating your pre-processed dataset and transforming it into meaningful insights or tools. This is your opportunity to be creative: choose techniques you find most appropriate and develop something that would genuinely help people compare artists and understand what makes them popular

You are encouraged to think on your own and develop something original. However, below are some suggestions or starting points:

- **Analytical deep dive:** Dive deeper into the factors the influence stream counts of artists and how they relate to each other (Career length vs solo/collaborative streams, genre vs total/solo/collaborative streams, etc.). Provide compelling visualizations that help highlight your discoveries.
- **Including New Variables:** Use the current set of variables to create new ones that are relevant and useful for the analysis. For example, an artist's career score calculated from the factors included in the data.
- **Insights Dashboard:** Combine the data you have with external sources and create an interactive dashboard that can be used to explore the streaming and artist insights.
- **Stream Count Prediction System:** Create a system that a user can use to predict the all-time stream count of a top artist from a set of relevant characteristics.

5. DELIVERABLES

5.1. Notebook submission

- A .ipynb notebook (or zip of multiple notebooks) featuring all the code you used throughout the project to:
  a. Decide on your final solutions for the problem at hand.
  b. Obtain your final results (code that helped you make decisions but didn't contribute directly to reaching the final solution should be included and commented).
  c. Any additional datasets required to run the notebook should be included in a zip file. (File naming: **GroupXX_PDS_2627.ipynb / GroupXX_PDS_2627.zip**).

5.2. Presentation

- A small 10-min free format presentation (e.g. PowerPoint with 10 slides, video, oral presentation) describing the main analytical processes and the conclusions obtained. **It will take place during the defense and must be delivered up to 2 days before it.**

File naming format: **GroupXX_PDS_2627**.ext (XX is your group number and .ext is the format of your presentation, such as .pptx, .pdf, .mp4, etc.)

The exact structure is flexible, but every question should be addressed in the presentation and part of the analysis. Notebook results must be properly explained.

6. Evaluation criteria and grading

Criteria	Description	Max Grade (Out of 20)
Data Wrangling and Analysis	Correct answers to the questions, as well as the ability to extract additional meaningful insights. And the overall quality of preprocessing and exploration.	6v
Advanced Topic	Level of success in completing the task. Originality, creativity, and effort in overcoming the challenges of the more advanced topics of the course.	6v
Data Science in Action	Relevancy of the proposed solution. Originality, depth, complexity and quality of execution affect the grade.	4v
Notebook Quality	Quality of the organization, explanation, and overall readability of the delivered notebook.	3v
Presentation	Effectiveness in conveying the insights extracted and the results of all the analysis performed. Mainly focused on the performance itself.	1v

- The notebook must begin with a Markdown cell listing the names and student numbers of all group members.
- Groups that include in the notebook a working, public GitHub repository link used during project development **will receive +0.5v**.
- Students can submit all deliverables with a **maximum delay of 3 days, incurring a penalty of 1v per day**. Beyond these three days, submissions will not be accepted.

Project Deadline: December 21st, 18h00

Presentations Deadline: 2 Days before the defense.

We will grade your projects based on the overall quality, correctness, clarity, and efficiency of your work. Below is a short general description of what is expected within each of the evaluation components.

1. **Data Wrangling & Analytics (6 points):** How well you understood and cleaned the issues present in the dataset. Additionally, we will evaluate how well you were able to extract insights from the data, namely:
    - Quality of preprocessing decisions, and their justifications, in accordance with what was required to rectify the issues they address.
    - Complete and correct answers to all the questions, with clear reasoning backing up how the answer was achieved.
    - Meaningfulness of the extracted insights, going beyond surface-level observations to find useful patterns in the streams of top artists.
2. **Advanced Topics (6 points):** This section evaluates how you were able to grasp the more advanced topics of the course when applied to a real-world situation. We'll look at:
    - The correctness of the implementation and the level of completion of the required tasks.
    - Whether your approach showcases that you truly understood these advanced concepts, rather than simply re-using and re-applying code.
    - The originality of your implementation. Did you add creative elements or improvements going beyond the required baseline.

3. **Data Science in Action (4 points):** This more open-ended section evaluates how well you managed your work to produce a relevant outcome. We'll take into consideration:
    - Pertinence and adequacy: does the final insight help interpret music streaming popularity?
    - How challenging or ambitious was your idea? Did you for a more complex idea or stick to basics?
    - The appropriateness of your methodology to the idea that you are aiming to develop.
    - How well you develop your idea and how you showcase its value.
    - Did you attempt to go outside of what was taught in class, such as external tools and data? Or did you use only the dataset provided and ideas provided?
4. **Notebook Quality (3 points):** A high-quality notebook is well-structured, easy to read, follow and understand even for someone reading it for the first time. What is expected:
    - Clear organization and logical sequencing of the steps performed.
    - Appropriate use of comments in code cells, supplemented by markdown cells for explaining the tasks being performed.
    - Readable code, which is straight to the point and efficient, avoiding unneeded repetition. At any given moment, your notebook should be clear about what you're doing, why, and what you extracted from doing it.
5. **Presentation (1 point):** This section focuses solely on the presentation during the defense.
    - We'll evaluate what you chose to include and how well you expressed the information you wanted to get across as well as how well you structured the overall presentation.
    - This component evaluates this presentation as performance, so the focus is primarily on what happens on the day of the defense.

7. Project Defense

- Project defenses will be in 40-minute slots (10-minute presentation + 30 minutes of technical defense).
- Defense should clearly explain:
    - The decisions that were made and why
    - How interactive/advanced components work
    - What is the importance of the final insight
- Project defense is **mandatory in person,** and **absence awards 0 for the project.**
- Different group members may have different evaluations based on their perceived contribution and performance during the defense.
- All group members **must understand the whole project.**
- The date of the defense will be defined during the semester.

8. FINAL REMARKS

- All topics mentioned will be evaluated based on the notebook - a well-structured and succinct notebook will have a big weight on the evaluation.
- If you use anything outside of the materials provided, it is crucial that you cite the source from which you got it.
- Before submitting, run your notebook from the start one last time, please comment out unnecessarily lengthy cells that take too much time to run. **Notebooks that don't run from start to finish will be penalized.**
- You should make sure to present the main conclusions of your work during the defense presentation.
- **Be extra careful with plagiarism and AI, as your work will be checked for these. Excessive use will result in penalties.**
- To avoid situations where we have conflicting versions, please make sure that you show, in the notebook, the version of the package you are using for each package you use.